Table 2: Performance comparison of different methods on next-item prediction. Bold scores are the best in each row, while underlined scores are the second best. Improvements over baselines are statistically significant with $p < 0.01$.

Datasets	Metric	POP	BPR-MF	NCF	FPMC	GRU4Rec	GRU4Rec$^+$	Caser	SASRec	BERT4Rec	Improv.
Beauty	HR@1	0.0077	0.0415	0.0407	0.0435	0.0402	0.0551	0.0475	0.0906	**0.0953**	5.19%
	HR@5	0.0392	0.1209	0.1305	0.1387	0.1315	0.1781	0.1625	0.1934	**0.2207**	14.12%
	HR@10	0.0762	0.1992	0.2142	0.2401	0.2343	0.2654	0.2590	0.2653	**0.3025**	14.02%
	NDCG@5	0.0230	0.0814	0.0855	0.0902	0.0812	0.1172	0.1050	0.1436	**0.1599**	11.35%
	NDCG@10	0.0349	0.1064	0.1124	0.1211	0.1074	0.1453	0.1360	0.1633	**0.1862**	14.02%
	MRR	0.0437	0.1006	0.1043	0.1056	0.1023	0.1299	0.1205	0.1536	**0.1701**	10.74%
Steam	HR@1	0.0159	0.0314	0.0246	0.0358	0.0574	0.0812	0.0495	0.0885	**0.0957**	8.14%
	HR@5	0.0805	0.1177	0.1203	0.1517	0.2171	0.2391	0.1766	0.2559	**0.2710**	5.90%
	HR@10	0.1389	0.1993	0.2169	0.2551	0.3313	0.3594	0.2870	0.3783	**0.4013**	6.08%
	NDCG@5	0.0477	0.0744	0.0717	0.0945	0.1370	0.1613	0.1131	0.1727	**0.1842**	6.66%
	NDCG@10	0.0665	0.1005	0.1026	0.1283	0.1802	0.2053	0.1484	0.2147	**0.2261**	5.31%
	MRR	0.0669	0.0942	0.0932	0.1139	0.1420	0.1757	0.1305	0.1874	**0.1949**	4.00%
ML-1m	HR@1	0.0141	0.0914	0.0397	0.1386	0.1583	0.2092	0.2194	0.2351	**0.2863**	21.78%
	HR@5	0.0715	0.2866	0.1932	0.4297	0.4673	0.5103	0.5353	0.5434	**0.5876**	8.13%
	HR@10	0.1358	0.4301	0.3477	0.5946	0.6207	0.6351	0.6692	0.6629	**0.6970**	4.15%
	NDCG@5	0.0416	0.1903	0.1146	0.2885	0.3196	0.3705	0.3832	0.3980	**0.4454**	11.91%
	NDCG@10	0.0621	0.2365	0.1640	0.3439	0.3627	0.4064	0.4268	0.4368	**0.4818**	10.32%
	MRR	0.0627	0.2009	0.1358	0.2891	0.3041	0.3462	0.3648	0.3790	**0.4254**	12.24%
ML-20m	HR@1	0.0221	0.0553	0.0231	0.1079	0.1459	0.2021	0.1232	0.2544	**0.3440**	35.22%
	HR@5	0.0805	0.2128	0.1358	0.3601	0.4657	0.5118	0.3804	0.5727	**0.6323**	10.41%
	HR@10	0.1378	0.3538	0.2922	0.5201	0.5844	0.6524	0.5427	0.7136	**0.7473**	4.72%
	NDCG@5	0.0511	0.1332	0.0771	0.2239	0.3090	0.3630	0.2538	0.4208	**0.4967**	18.04%
	NDCG@10	0.0695	0.1786	0.1271	0.2895	0.3637	0.4087	0.3062	0.4665	**0.5340**	14.47%
	MRR	0.0709	0.1503	0.1072	0.2273	0.2967	0.3476	0.2529	0.4026	**0.4785**	18.85%

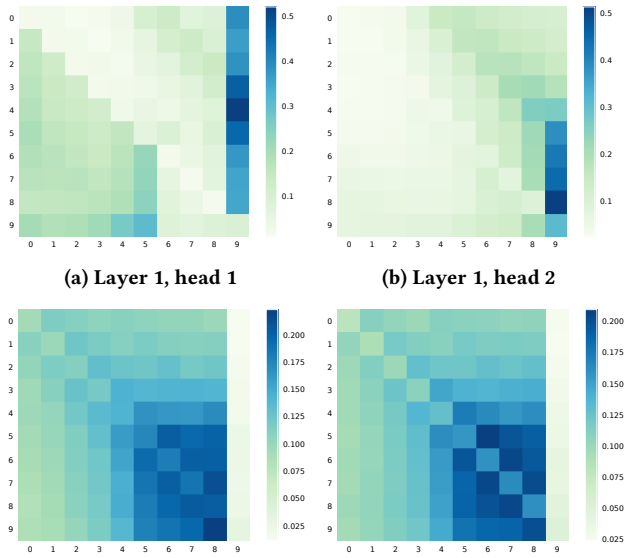
Table 3: Analysis on bidirection and Cloze with $d = 256$.

Model	Beauty			ML-1m		
	HR@10	NDCG@10	MRR	HR@10	NDCG@10	MRR
SASRec	0.2653	0.1633	0.1536	0.6629	0.4368	0.3790
BERT4Rec (1 mask)	0.2940	0.1769	0.1618	0.6869	0.4696	0.4127
BERT4Rec	0.3025	0.1862	0.1701	0.6970	0.4818	0.4254

To answer this question, we try to isolate the effects of these two factors by constraining the Cloze task to mask only one item at a time. In this way, the main difference between our BERT4Rec (with 1 mask) and SASRec is that BERT4Rec predicts the target item jointly conditioning on both left and right context. We report the results on Beauty and ML-1m with $d = 256$ in Table 3 due to the space limitation. The results show that BERT4Rec with 1 mask significantly outperforms SASRec on all metrics. It demonstrates the importance of bidirectional representations for sequential recommendation. Besides, the last two rows indicate that the Cloze objective also improves the performances. Detailed analysis of the mask proportion ρ in Cloze task can be found in § 4.6

Question 2: *Why and how does bidirectional model outperform unidirectional models?*

To answer this question, we try to reveal meaningful patterns by visualizing the average attention weights of the last 10 items during the test on Beauty in Figure 2. Due to the space limitation, we only report four representative attention heat-maps in different layers and heads.



(a) Layer 1, head 1 **(b) Layer 1, head 2**

(c) Layer 2, head 2 **(d) Layer 2, head 4**

Figure 2: Heat-maps of average attention weights on Beauty, the last position "9" denotes "`[mask]`" (best viewed in color).

We make several observations from the results. ***a)*** Attention varies across different heads. For example, in layer 1, head 1 tends

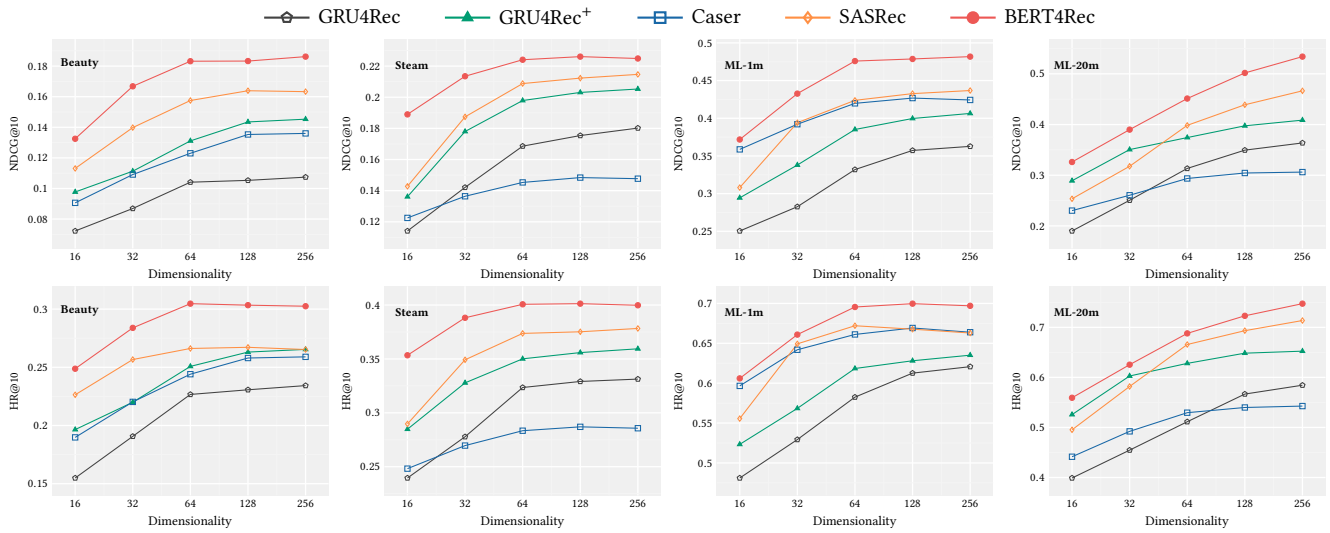


Figure 3: Effect of the hidden dimensionality d on HR@10 and NDCG@10 for neural sequential models.

to attend on items at the left side while head 2 prefers to attend on items on the right side. ***b***) Attention varies across different layers. Apparently, attentions in layer 2 tend to focus on more recent items. This is because layer 2 is directly connected to the output layer and the recent items play a more important role in predicting the future. Another interesting pattern is that heads in Figure 2a and 2b also tend to attend on `[mask]`[13]. It may be a way for self-attention to propagate sequence-level state to the item level. ***c***) Finally and most importantly, unlike unidirectional model can only attend on items at the left side, items in BERT4Rec tend to attend on the items at both sides. This indicates that bidirectional is essential and beneficial for user behavior sequence modeling.

In the following studies, we examine the impact of the hyper-parameters, including the hidden dimensionality d, the mask proportion ρ, and the maximum sequence length N. We analyze one hyper-parameter at a time by fixing the remaining hyper-parameters at their optimal settings. Due to space limitation, we only report NDCG@10 and HR@10 for the follow-up experiments.

4.5 Impact of Hidden Dimensionality d

We now study how the hidden dimensionality d affects the recommendation performance. Figure 3 shows NDCG@10 and HR@10 for neural sequential methods with the hidden dimensionality d varying from 16 to 256 while keeping other optimal hyper-parameters unchanged. We make some observations from this figure.

The most obvious observation from these sub-figures is that the performance of each model tends to converge as the dimensionality increases. A larger hidden dimensionality does not necessarily lead to better model performance, especially on sparse datasets like Beauty and Steam. This is probably caused by overfitting. In terms of details, Caser performs unstably on four datasets, which might limit its usefulness. Self-attention based methods (*i.e.*, SASRec and BERT4Rec) achieve superior performances on all datasets. Finally, our model consistently outperforms all other baselines on all datasets even with a relatively small hidden dimensionality. Considering that our model achieves satisfactory performance with $d \geq 64$, we only report the results with d=64 in the following analysis.

[13]This phenomenon also exists in text sequence modeling using BERT.

4.6 Impact of Mask Proportion ρ

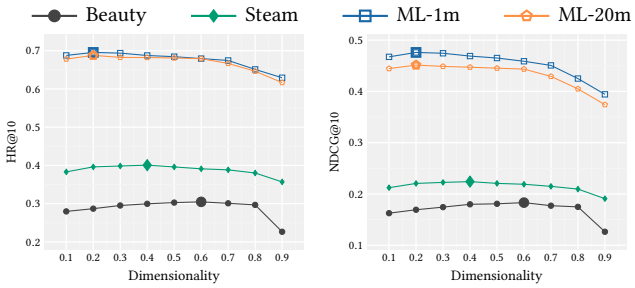


Figure 4: Performance with different mask proportion ρ on $d = 64$. Bold symbols denote the best scores in each line.

As described in § 3.6, mask proportion ρ is a key factor in model training, which directly affects the loss function (Equation 8). Obviously, mask proportion ρ should not be too small or it is not enough to learn a strong model. Meanwhile, it should not be too large, otherwise, it would be hard to train since there are too many items to guess based on a few contexts in such case. To examine this, we study how mask proportion ρ affects the recommendation performances on different datasets.

Figure 4 shows the results with varying mask proportion ρ from 0.1 to 0.9. Considering the results with $\rho > 0.6$ on all datasets, a general pattern emerges, the performances decreasing as ρ increases. From the results of the first two columns, it is easy to see that $\rho = 0.2$ performs better than $\rho = 0.1$ on all datasets. These results verify what we claimed above.

In addition, we observe that the optimal ρ is highly dependent on the sequence length of the dataset. For the datasets with short sequence length (*e.g.*, Beauty and Steam), the best performances are

Table 4: Performance with different maximum length N.

		10	20	30	40	50
Beauty	#samples/s	5504	3256	2284	1776	1441
	HR@10	0.3006	0.3061	0.3057	0.3054	0.3047
	NDCG@10	0.1826	0.1875	0.1837	0.1833	0.1832
		10	50	100	200	400
ML-1m	#samples/s	14255	8890	5711	2918	1213
	HR@10	0.6788	0.6854	0.6947	0.6955	0.6898
	NDCG@10	0.4631	0.4743	0.4758	0.4759	0.4715

achieved at ρ=0.6 (Beauty) and ρ=0.4 (Steam), while the datasets with long sequence length (*e.g.*, ML-1m and ML-20m) prefer a small ρ=0.2. This is reasonable since, compared with short sequence datasets, a large ρ in long sequence datasets means much more items that need to be predicted. Take ML-1m and Beauty as example, ρ=0.6 means we need to predict 98=$\lfloor 163.5 \times 0.6 \rfloor$ items on average per sequence for ML-1m, while it is only 5=$\lfloor 8.8 \times 0.6 \rfloor$ items for Beauty. The former is too hard for model training.

4.7 Impact of Maximum Sequence Length N

We also investigate the effect of the maximum sequence length N on model's recommendation performances and efficiency.

Table 4 shows recommendation performances and training speed with different maximum length N on Beauty and ML-1m. We observe that the proper maximum length N is also highly dependent on the average sequence length of the dataset. Beauty prefers a smaller $N = 20$, while ML-1m achieves the best performances on $N = 200$. This indicates that a user's behavior is affected by more recent items on short sequence datasets and less recent items for long sequence datasets. The model does not consistently benefit from a larger N since a larger N tends to introduce both extra information and more noise. However, our model performs very stably as the length N becomes larger. This indicates that our model can attend to the informative items from the noisy historical records.

A scalability concern about BERT4Rec is that its computational complexity per layer is $O(n^2d)$, quadratic with the length n. Fortunately, the results in Table 4 shows that the self-attention layer can be effectively parallelized using GPUs.

4.8 Ablation Study

Finally, we perform ablation experiments over a number of key components of BERT4Rec in order to better understand their impacts, including positional embedding (PE), position-wise feed-forward network (PFFN), layer normalization (LN), residual connection (RC), dropout, the layer number L of self-attention, and the number of heads h in multi-head attention. Table 5 shows the results of our default version ($L = 2, h = 2$) and its eleven variants on all four datasets with dimensionality $d = 64$ while keeping other hyper-parameters (*e.g.*, ρ) at their optimal settings.

We introduce the variants and analyze their effects respectively:

(1) **PE**. The results show that removing positional embeddings causes BERT4Rec's performances decreasing dramatically on long sequence datasets (*i.e.*, ML-1m and ML-20m). Without the positional embeddings, the hidden representation $\boldsymbol{H}_i^L$ for each item v_i depends only on item embeddings. In this situation, we predict different target items using the same hidden representation of "`[mask]`". This makes the model ill-posed. This issue is more serious on long sequence datasets since they have more masked items to predict.

(2) **PFFN**. The results show that long sequence datasets (*e.g.*, ML-20m) benefit more from PFFN. This is reasonable since a purpose of PFFN is to integrate information from many heads which are preferred by long sequence datasets as discussed in the analysis about head number h in ablation study (5).

(3) **LN**, **RC**, and **Dropout**. These components are introduced mainly to alleviate overfitting. Obviously, they are more effective on small datasets like Beauty. To verify their effectiveness on large datasets, we conduct an experiment on ML-20m with layer L=4. The results show that NDCG@10 decreases about 10% w/o RC.

(4) **Number of layers** L. The results show that stacking Transformer layer can boost performances especially on large datasets (*e.g*, ML-20m). This verifies that it is helpful to learn more complex item transition patterns via deep self-attention architecture. The decline in Beauty with $L = 4$ is largely due to overfitting.

(5) **Head number** h. We observe that long sequence datasets (*e.g.*, ML-20m) benefit from a larger h while short sequence datasets (*e.g.*, Beauty) prefer a smaller h. This phenomenon is consistent with the empirical result in [48] that large h is essential for capturing long distance dependencies with multi-head self-attention.

Table 5: Ablation analysis (NDCG@10) on four datasets. Bold score indicates performance better than the default version, while ↓ indicates performance drop more than 10%.

Architecture	Dataset			
	Beauty	Steam	ML-1m	ML-20m
$L = 2, h = 2$	0.1832	0.2241	0.4759	0.4513
w/o PE	0.1741	0.2060	0.2155↓	0.2867↓
w/o PFFN	0.1803	0.2137	0.4544	0.4296
w/o LN	0.1642↓	0.2058	0.4334	0.4186
w/o RC	0.1619↓	0.2193	0.4643	0.4483
w/o Dropout	0.1658	0.2185	0.4553	0.4471
1 layer ($L = 1$)	0.1782	0.2122	0.4412	0.4238
3 layers ($L = 3$)	**0.1859**	**0.2262**	**0.4864**	**0.4661**
4 layers ($L = 4$)	**0.1834**	**0.2279**	**0.4898**	**0.4732**
1 head ($h = 1$)	**0.1853**	0.2187	0.4568	0.4402
4 heads ($h = 4$)	0.1830	**0.2245**	**0.4770**	**0.4520**
8 heads ($h = 8$)	0.1823	**0.2248**	0.4743	**0.4550**

5 CONCLUSION AND FUTURE WORK

Deep bidirectional self-attention architecture has achieved tremendous success in language understanding. In this paper, we introduce a deep bidirectional sequential model called BERT4Rec for sequential recommendation. For model training, we introduce the Cloze task which predicts the masked items using both left and right